HARVARD COLLEGE
Office of Undergraduate Research and Fellowships

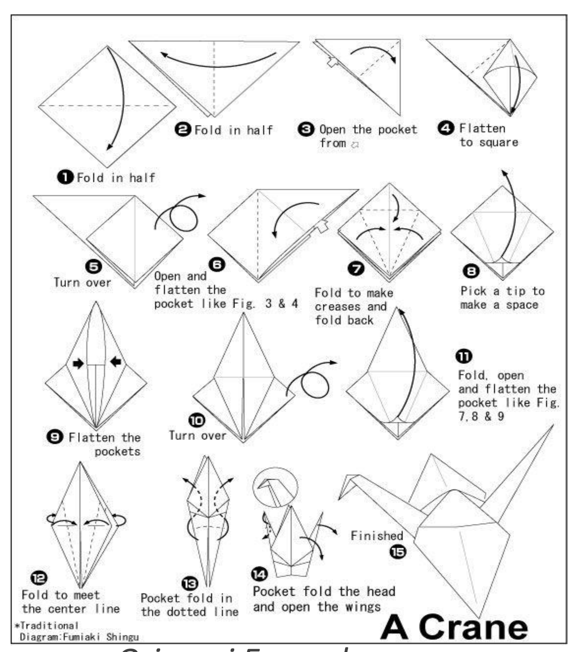
Exploring Textile Foldability For Origami-Inspired Structures

Faith Atieno, Kausalya Mahadevan, Bertoldi Group

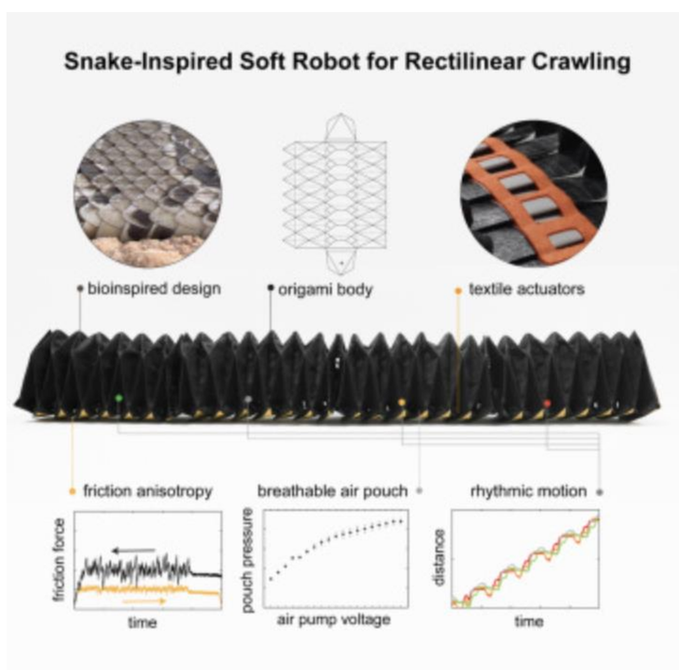
PRISE
PROGRAM FOR RESEARCH IN SCIENCE AND ENGINEERING

Introduction

Textiles are promising for deployable structures due to their flexibility, light weight, and everyday use.
Flat sheets can turn into 3D forms, often inspired by origami folding
Here, the study explores how stitch patterns and fabric properties affect textile folding when actuated by heat-shrinking and water soluble thread.

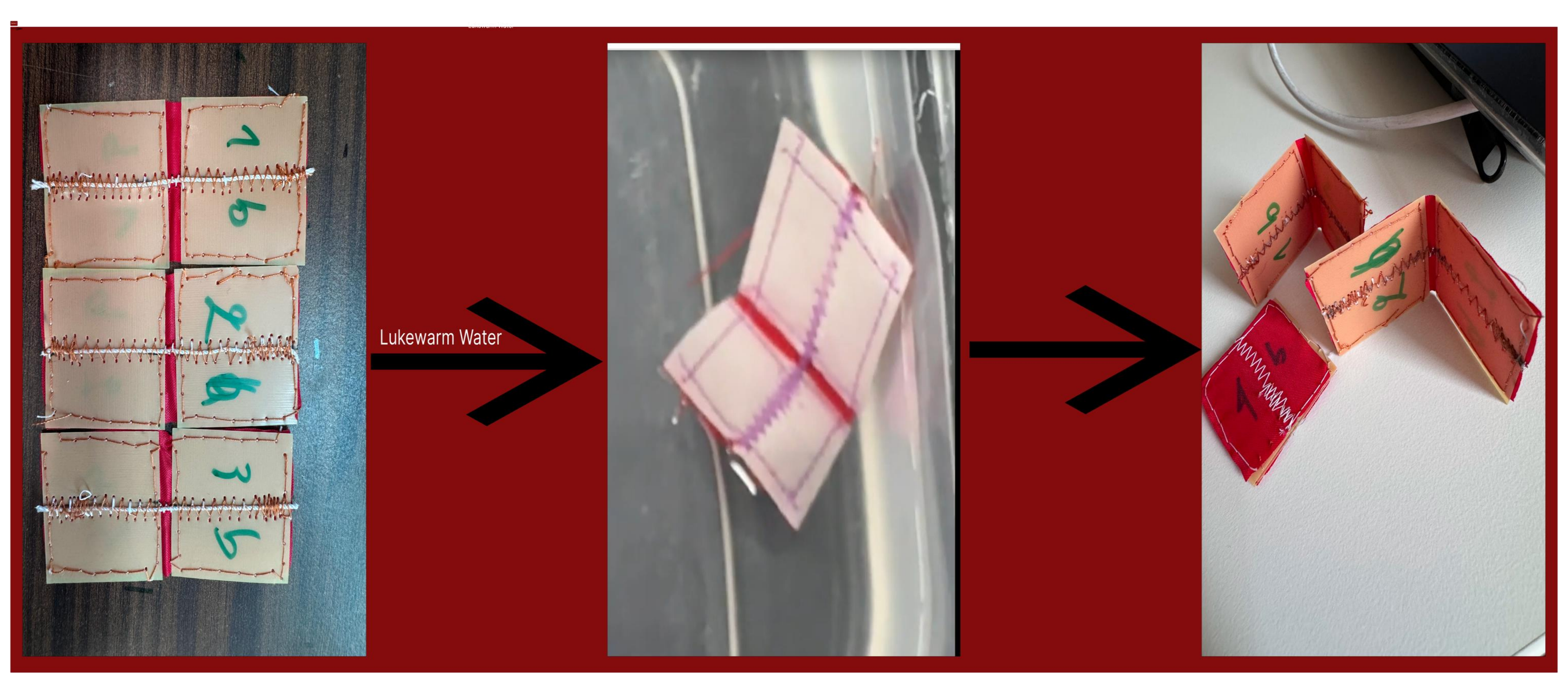


Origami Example



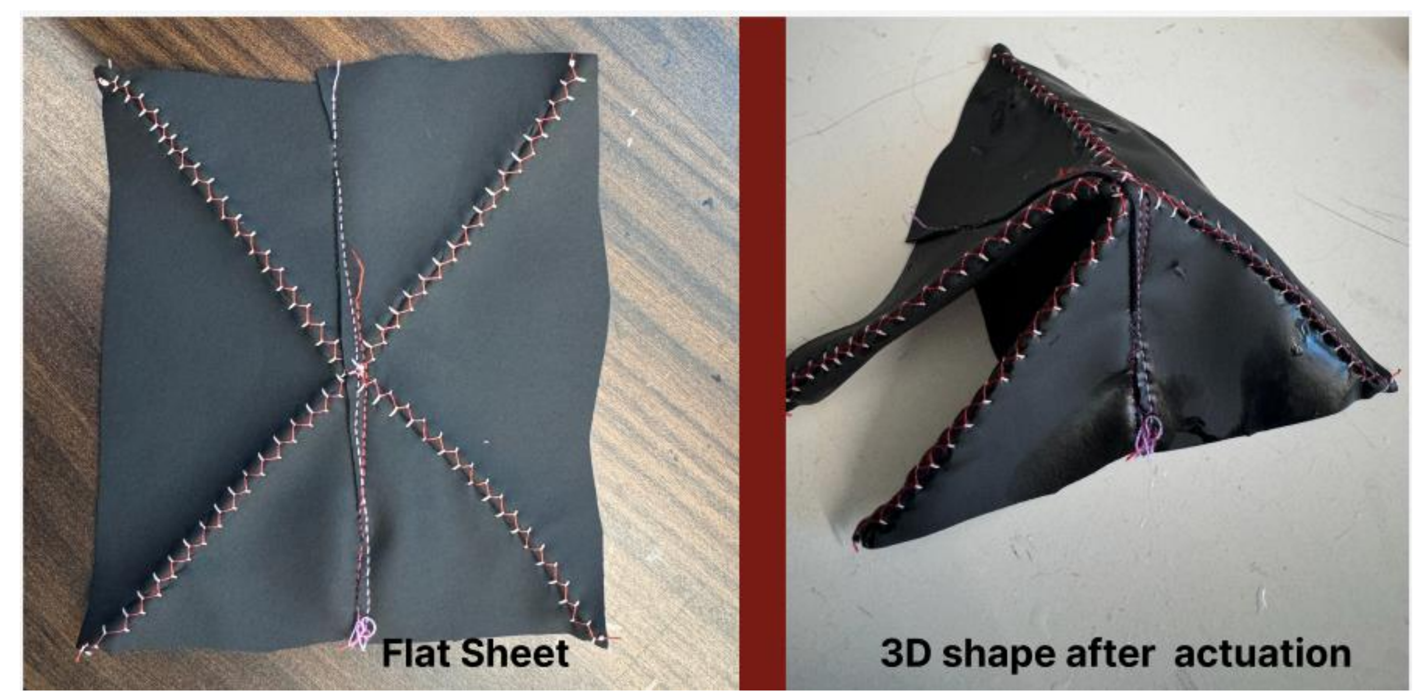
Methods

We stitched actuators into different fabrics (chiffon, woven nylon, and quilting cotton)using different patterns
We varied the fabric type and the distance between panels to observe differences in folding behavior for the same stitch pattern.
Immersed the different samples in activation environments to observe behavior and move on from there
Used resulting behavior to inform decisions on panel spacing, material selection, and stitch design

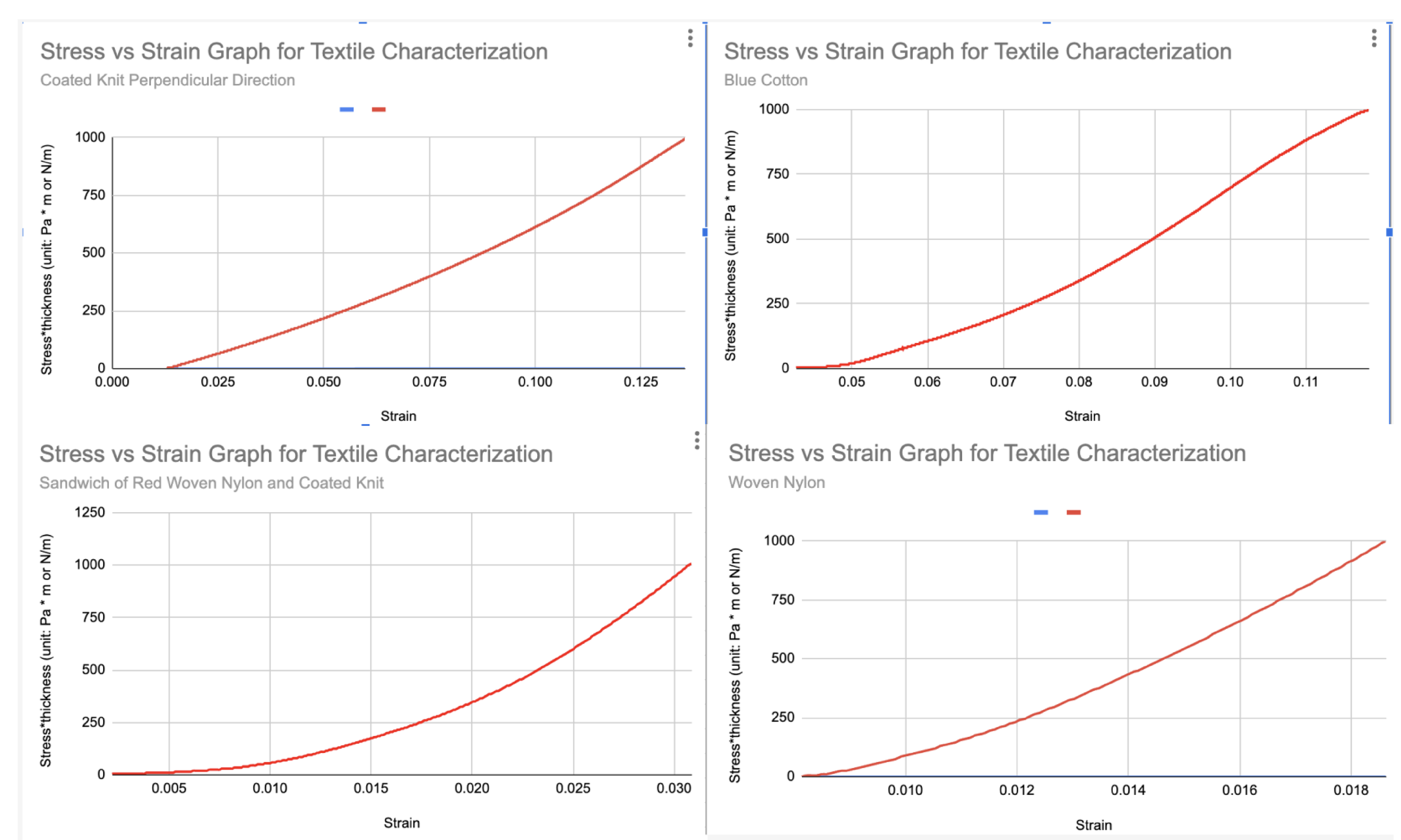


Results

Chiffon folded easily, while woven nylon resisted tight folds but still held its transformed shape well. Quilting cotton showed an in-between performance. The behaviors opened possibilities of combining materials to get the folding or output we want.
For 7 cm-wide woven-nylon samples, spacing of 5–10 mm between panels led to complete folding with near-zero-degree angles. Spacing <5 mm resulted in partial folding with acute angles.
The same stitch pattern produced different results on different fabrics, highlighting the importance of material choice in fold predictability.
Mechanical properties such as Young's modulus, permeability, and foldability also affected performance.



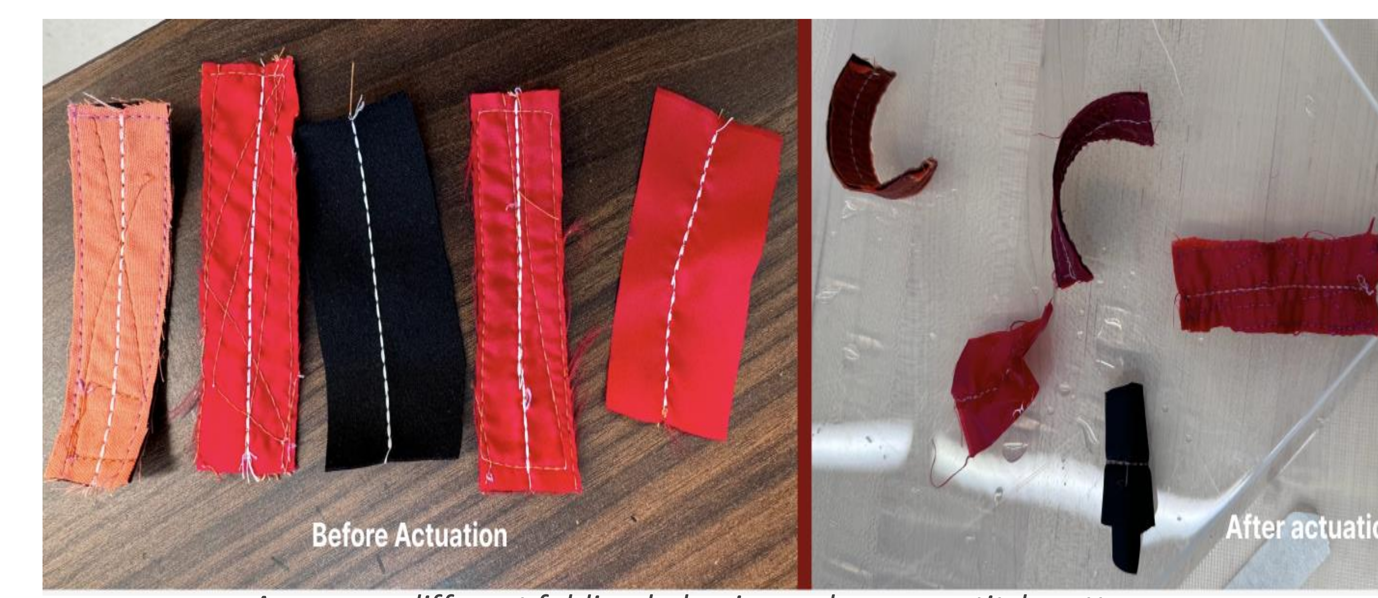
Mechanical properties such as Young's modulus, permeability, and thickness also affected performance



Different used materials' Stress vs Strain graphs

Conclusion

Stitch pattern and fabric type both play key roles in how textiles fold.
Material stiffness and structure affect how well the folds form and whether they hold their shape.
Panel spacing controls the fold angle and can determine whether the fold completes or not.
These findings are useful for deployable structures and can help guide decisions on what stitch patterns and fabric types to use when designing 2D textiles for specific 3D shapes.
This work also shows how understanding material behavior can help us create custom designs that fold more reliably when actuated.

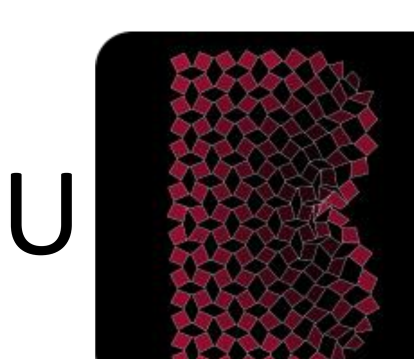


A snap on different folding behavior under same stitch pattern

Next Steps

Use insights from material performance (e.g. stiffness, flexibility) to design or combine custom fabrics that optimize folding.
Investigate how to better control the transition to achieve more stable and predictable 3D shapes.
Begin developing a simple model (computational or rule-based) to predict foldability based on fabric properties and stitch settings.

THANK YOU



Acknowledgements
This work was supported by a PRISE award from the Harvard College Office of Undergraduate Research and Fellowships.

References
Seyidoğlu, B., & Rafsanjani, A. (2024). *A textile origami snake robot for rectilinear locomotion*. *Device*, 2(2).
Farhadi, D., Wang, H., Jiang, H., & Rafsanjani, A. (2025). *Origami crawlers: Exploring a single origami vertex for complex path navigation*. *Advanced Materials*, Article 2502293.
Chang, Z., Noma, Y., Feng, S., Yang, X., Shinoda, K., Ta, T. D., ... & Roumen, T. (2024). *OriStitch: A machine embroidery workflow to turn existing fabrics into self-folding 3D textiles*. *arXiv preprint* arXiv:2412.02891.

faithatieno@college.harvard.edu/+18572704280
faithannebelle@gmail.com